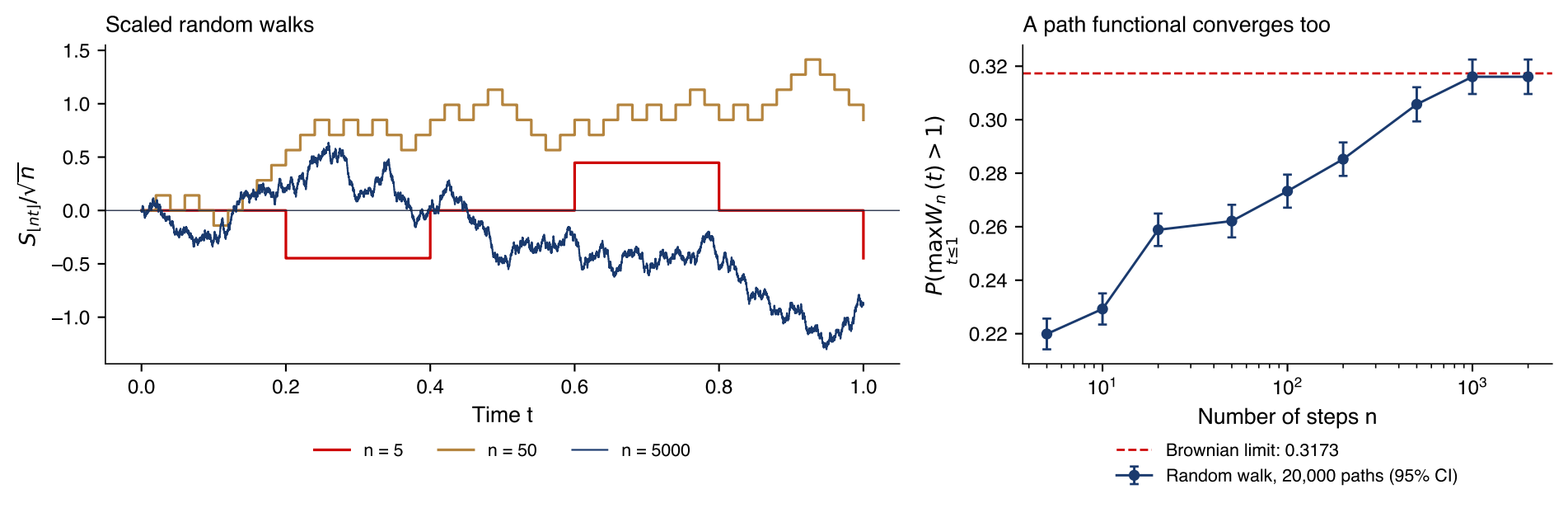

Scaled random walks
$S_{\lfloor nt \rfloor}/\sqrt{n}$
1.5
1.0
0.5
0.0
−0.5
−1.0
0.0
0.2
0.4
0.6
0.8
1.0
Time t
n = 5
n = 50
n = 5000
A path functional converges too
$P(\max_{t \le 1} W_n(t) > 1)$
0.32
0.30
0.28
0.26
0.24
0.22
10^1
10^2
10^3
Number of steps n
Brownian limit: 0.3173
Random walk, 20,000 paths (95% CI)